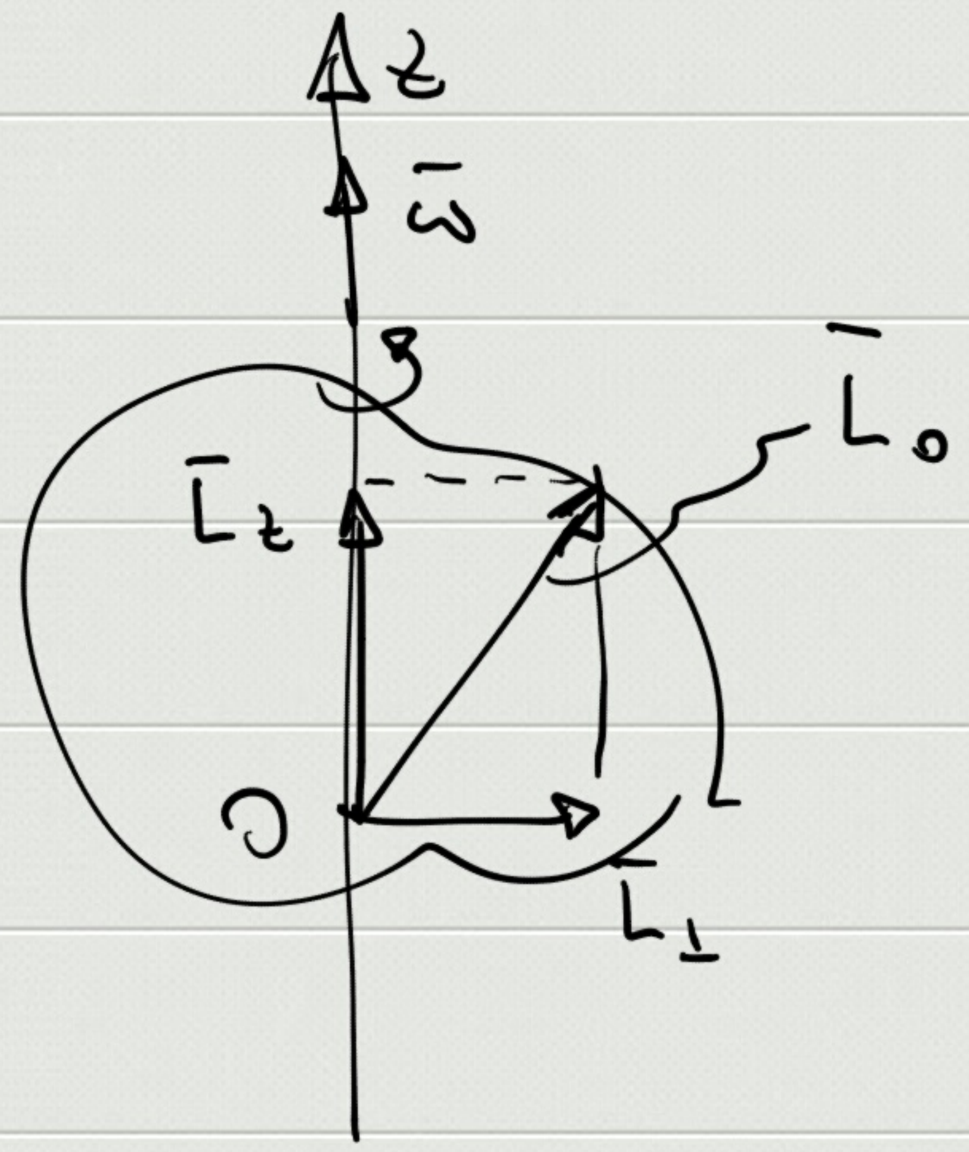
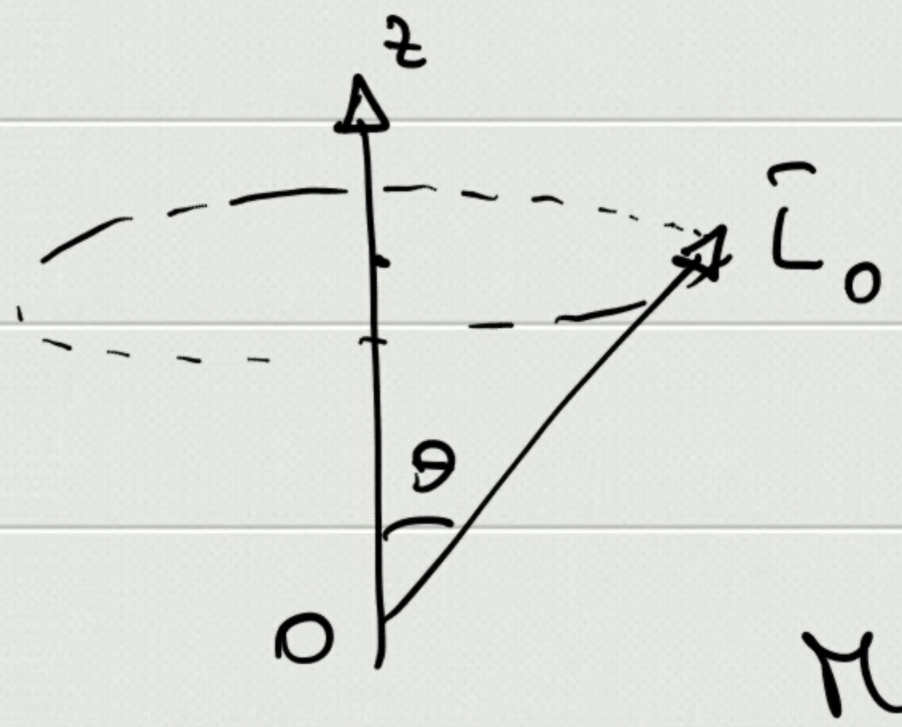


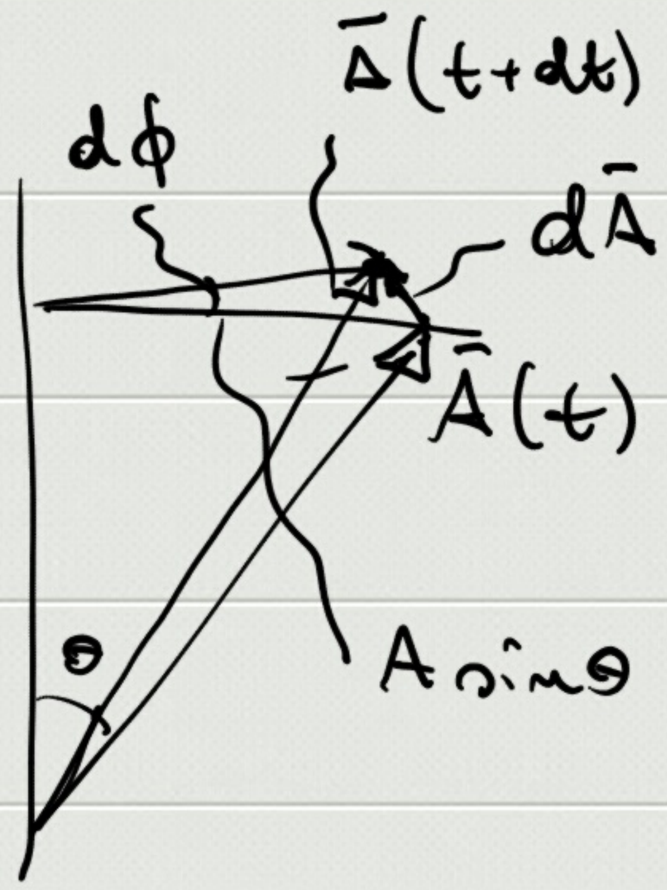
$$\bar{L}_0 = L_{0,z}\,\bar{u}_z + L_{0,\perp}\,\bar{u}_\perp$$



$\omega = \text{cost}$

$\Rightarrow |\bar{L}_0| = \text{cost}$

Moto di precessione



$$dA = A\sin\theta\, d\phi$$

$$\frac{dA}{dt} = A\sin\theta\,\frac{d\phi}{dt} =$$

$$= A\omega\sin\theta$$

$$\boxed{\frac{d\bar{A}}{dt} = \bar{\omega}\times\bar{A}}$$

$\bar{L}_0$: moto precessione $\bar{\omega} = \text{cost} \Rightarrow$

$$\frac{d\bar{L}_0}{dt} = \bar{\omega} \times \bar{L} \Rightarrow \left| \frac{d\bar{L}_0}{dt} \right| = |\bar{\omega} \times \bar{L}| = \omega L_\perp$$

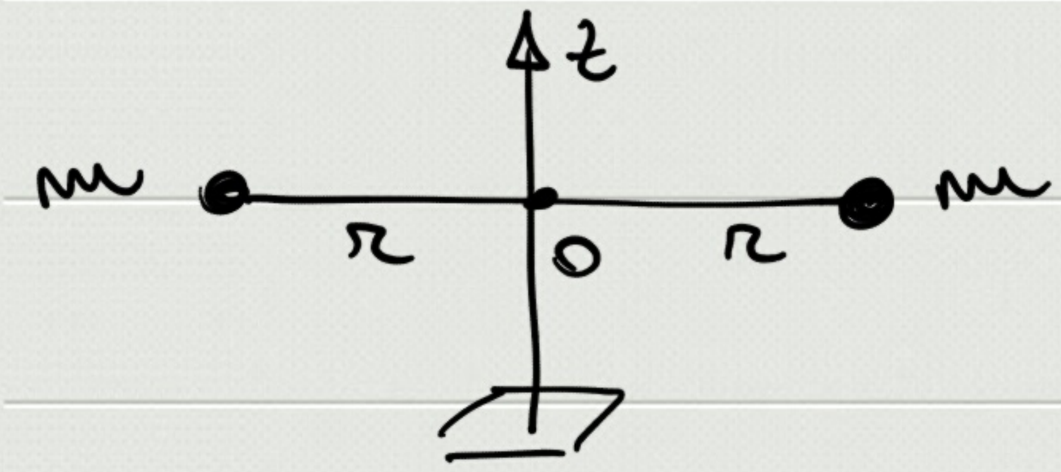
$$= \bar{M}_0^E - \underbrace{\cancel{\bar{v}_0 \times m \bar{v}_{cm}}}_{=0}$$

$$\boxed{\bar{M}_0^E = \omega L_\perp}$$

z è asse princ. inerzia $\Rightarrow L_\perp = 0$

z non è asse princ. inerzia $\Rightarrow L_\perp \neq 0$

$$\Rightarrow \boxed{\bar{M}_0^E \neq 0}$$



$\sum_i m_i R_i^2$

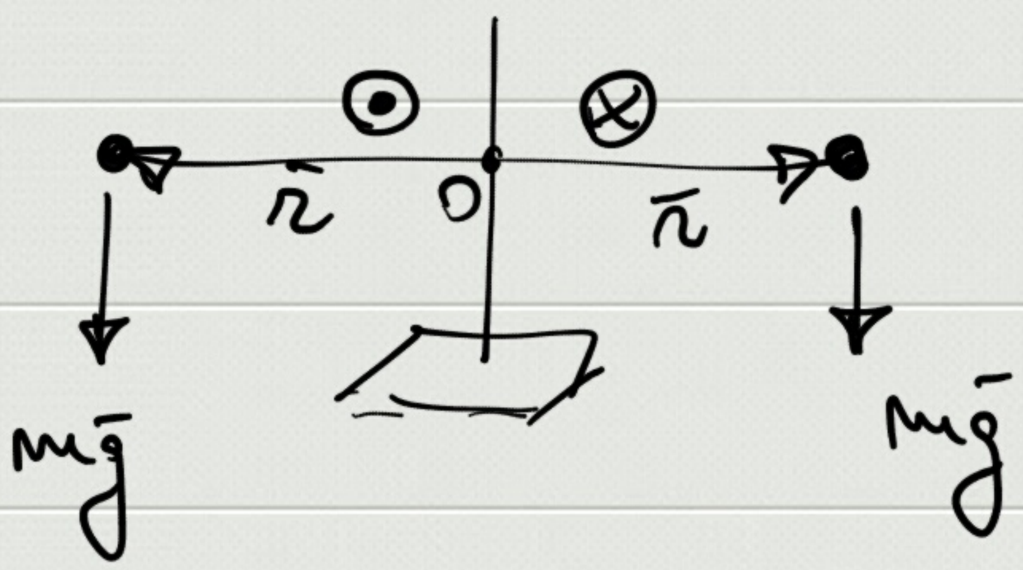
$I_z = 2mr^2$

$\Rightarrow L_z = 2mr^2\omega$

$L_\perp = 0$

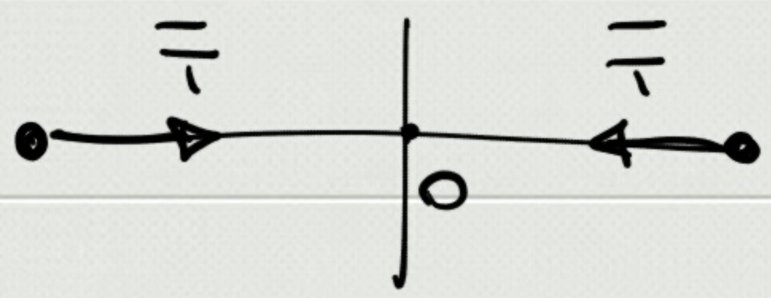
$\Rightarrow \bar{M}_O^E = 0$

$$\frac{d\bar{L}_O}{dt} = \bar{M}_O^E = 0 \Rightarrow \boxed{\bar{L}_O = \text{cost}}$$

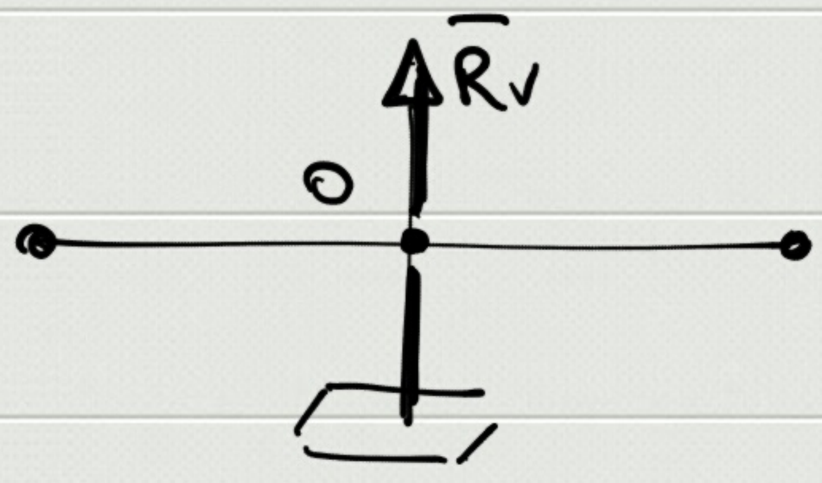


$\bar{M}_O = \bar{r} \times \bar{F}$

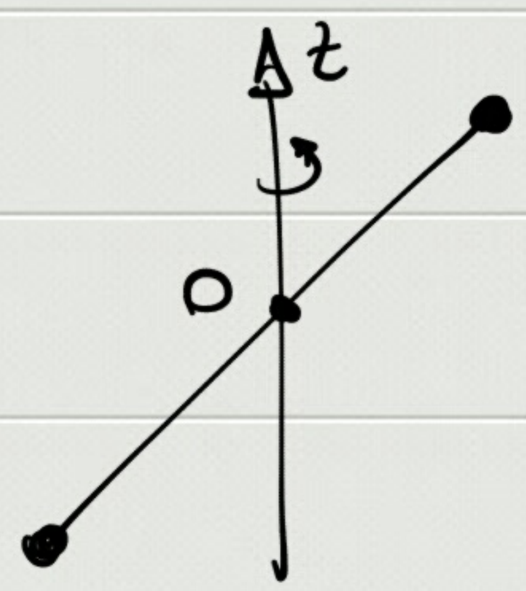
$\bar{M}_{peso} = 0$



$\bar{M}_{tensioni} = 0$



$\bar{M}_{vincolare} = 0$



$$M_O^{\varepsilon} = \omega L_{\perp}$$